3.3.12 Polymers (A-level only)

The study of polymers is extended to include condensation polymers. The ways in which condensation polymers are formed are studied, together with their properties and typical uses. Problems associated with the reuse or disposal of both addition and condensation polymers are considered.

3.3.12.1 Condensation polymers (A-level only)

Content	Opportunities for skills development
Condensation polymers are formed by reactions between: • dicarboxylic acids and diols • dicarboxylic acids and diamines • amino acids. The repeating units in polyesters (eg Terylene) and polyamides (eg nylon 6,6 and Kevlar) and the linkages between these repeating units. Typical uses of these polymers. **Students should be able to:** • draw the repeating unit from monomer structure(s) • draw the repeating unit from a section of the polymer chain • draw the structure(s) of the monomer(s) from a section of the polymer • explain the nature of the intermolecular forces between molecules of condensation polymers.	**AT k** **PS 1.2** Making nylon 6,6

3.3.12.2 Biodegradability and disposal of polymers (A-level only)

Content	Opportunities for skills development
Polyalkenes are chemically inert and non-biodegradable. Polyesters and polyamides can be broken down by hydrolysis and are biodegradable. The advantages and disadvantages of different methods of disposal of polymers, including recycling. **Students should be able to:** • explain why polyesters and polyamides can be hydrolysed but polyalkenes cannot.	**Research opportunity** Students could research problems associated with the disposal of different polymers.